SnV Diamond Nanobeam Cavities: Process-Aware and Crystallography-Aware Design

Simulation workflow and implementation plan. Created 2026-10-16; revised 2026-10-17. This revision incorporates the technical review: explicit pre-undercut protection geometry, etch-law limits, emitter linewidth and tuning dynamics, leakage extraction, polarization bounds, fair controls and decision gates.

Overview

Build **one shared simulation framework**, then use it to test two distinct hypotheses:

- **Idea 1: process-aware cavity design.** Can the mask and release layout be optimized so that *the structures produced by quasi-isotropic etching* keep good optical confinement and useful SnV coupling?
- **Idea 2: crystallography-aware field design.** Can local cavity modifications improve coupling to a chosen SnV dipole *beyond what simply rotating a conventional cavity achieves*, while staying compatible with the release process?

Idea 1 is the main research project. Idea 2 begins as a short, quantitative feasibility study using the same stored fields. Expand Idea 2 only after it demonstrates a useful-emission advantage over the best rotated conventional cavity under the same fabrication and emitter assumptions. The first milestone is to identify a specific release-induced optical failure before choosing the optimization objective.

Starting software stack (details in Section 6):

$$\text{COMSOL Wave Optics} + \text{MATLAB automation} + \text{Python analysis/optimization} + \text{GDS layout tools.}$$

Add **one independent electromagnetic solver later**, not several at the start. The largest piece of custom development will be the **etch-to-geometry model**, not the Maxwell solver.

Project status: This is an implementation specification. It does not report completed COMSOL models, validated etch parameters, simulation results or measured yield. First freeze the baseline in Section 1.3; begin optimization only after the gates in Section 7 pass.

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1. Shared foundation for both projects

1.1 Fix the scope before building an optimizer

For the first study, use a **one-dimensional, mainly TE-like diamond nanobeam cavity near 620 nm** with one designated output waveguide. Start with a conventional hole-based cavity. Add an open-corrugation (Sawfish-type) comparator once the pipeline works.

The Chen et al. device [1] is a relevant starting point for bulk-undercut fabrication, and Sawfish [11] is a strong benchmark for collection-oriented design. Existing work on optimizing for fabrication error should inform the comparison. The new paper should not present it as something it introduces.

Item	Initial choice
Optical target	A selected SnV transition near 620 nm
Structure	Suspended diamond nanobeam with a tapered photonic-crystal defect
Output	One specified guided mode at a defined reference plane
Fabrication	Transfer to d_1, protection, breakthrough, extra directional etch d_2, release
Starting thickness	Reproduce the published baseline first; then investigate your achievable thickness range
Crystal coordinates	(001) as the reference plane for a $\{100\}$ wafer; enter the actual wafer rotation/miscut separately
Material model	Scalar diamond permittivity at first; include dispersion and material uncertainty in final calculations
Main endpoint	Useful emission probability or a clearly defined coupling metric, not nominal Q alone

A published refractive-index model is fine for early simulations. Its uncertainty still matters for any claim about absolute wavelength targeting. Turri et al. [2] give measured dispersion data from the visible to the near-infrared.

Start the benchmark with a declared constant index. When introducing $n(\omega)$, the eigenproblem becomes nonlinear: use a validated nonlinear eigenfrequency method or iterate the material evaluation frequency self-consistently until the same tracked mode and frequency converge below the declared spectral tolerance [36]. Store the material evaluation frequency and residual; a single index evaluation at an unrelated nominal frequency is not a converged dispersive eigenmode.

> **Do not start with simultaneous optimization of optics, phononics, spin coherence, implantation and mechanical survival.** These are optical-design papers. Other physics should enter only where it changes the optical conclusion.

1.2 Build the software around separate modules

The central pipeline has two connected stages:

1. **Geometry:** mask and release layout → fabrication model → 3D geometry.
2. **Evaluation:** 3D geometry → Maxwell solution → emitter/output metrics → optimization.

The optimizer proposes a new mask/layout and repeats both stages. Geometry validation runs before every Maxwell solve.

Keep these modules separate:

Module	Input	Output
Mask generator	Hole dimensions, periods, taper coefficients, beam width	Fabricable 2D contours
Process model	Mask, release layout, process variables	3D diamond geometry
Geometry validator	Generated solid	Connectivity, feature sizes, release status
Electromagnetic solver	Geometry and material model	Bands, resonances, fields, losses
Emitter model	Fields, dipoles, positions, transition parameters	Coupling and emission probabilities
Statistical evaluator	Process and emitter distributions	Yield, quantiles, uncertainty intervals
Optimizer	Design variables and objectives	Candidate masks and layouts

This separation is essential. **The optimizer should control variables you can set on the mask. It should not freely edit a final 3D structure that your fabrication process cannot produce.**

COMSOL Wave Optics is the main solver: it handles FEM calculations on flexible geometry and supports eigenfrequency and Floquet-periodic analyses [3]. Use a second formulation, such as Meep, to cross-check selected cases; it has documented workflows for cavities, LDOS and mode decomposition [4-6]. Section 6 says which program owns each task and how data move between them.

1.3 Freeze a traceable baseline

Selected reference: the hole-based, Gaussian-chirped cavity in Chen et al. [1,9]. Start with the versioned preprint [9], Section 2 and Fig. 2; check the design against the published article and Supplement 1 before declaring a reproduction. Keep the isolated-cavity reproduction separate from the later output-coupled design.

The main text reports these anchors [9]:

Reported item	Value	Correct use
Measured beam width	260 nm	Fabricated-geometry reference
Measured thickness	201 nm	Fabricated-geometry reference
Measured hole radius	64 nm	Fabricated-geometry reference
Simulated fundamental-mode quality factor	3×10^6	Match simulated geometry and loss channels
Simulated mode volume	$0.8(\lambda/n)^3$	Match the volume definition

The measured dimensions do not completely specify the simulated cavity. Do not combine them with an invented chirp and call it a reproduction. Experimental and ideal simulated Q are separate comparisons.

Create `baseline_spec.json` with each parameter's value, unit, source location and evidence status: `reported`, `digitized`, `assumed` or `unresolved`. Resolve:

- Mirror/central periods, exact chirp law, hole shape and radius variation.
- Taper and left/right mirror cell counts, center indexing and hole positions.

- Simulated cross section, material model, surroundings, supports and substrate gap.
- Output termination, reference plane, guided mode and loaded/intrinsic Q definition.
- Reference resonance, mode-volume definition and available source solver settings.

Assign uncertainty to digitized values. Reject unresolved essential inputs instead of substituting undocumented defaults. A fully specified provisional geometry may test the software, but must be labeled as a development case and cannot pass the reproduction gate.

Alternative validation route: if recovering the Chen chirp, periods or cell counts blocks G0, select another fully specified published cavity and freeze its complete source-traceable specification. The rectangular cavity of Mouradian et al. [31] is a candidate; verify center indexing, terminations and material assumptions before using it. Reproduce that reference at its original wavelength first, then label any adaptation to 620 nm as a new design. Retain Chen as a fabrication comparator. This route passes the same numerical gates and never turns missing dimensions into a claimed reproduction.

Deliverable: frozen parameter file, provenance table, geometry preview and comparison observables. Archive the source version, hash the specification and create a new version whenever a parameter changes.

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2. The simulations both papers need

2.1 Simulation A: cross-section modes of a straight waveguide

Purpose: Rule out unsuitable thickness, width and cross-section combinations before building full cavities.

Solve a 2D transverse eigenmode problem at about 620 nm for several cross sections:

- Ideal rectangle
- Slanted sidewalls
- Faceted or curved underside
- Laterally asymmetric underside

Extract the effective indices and vector fields of the fundamental mode and nearby higher-order modes.

This is a screening calculation. It shows whether a geometry supports the desired guided mode and whether another polarization family sits nearby. **It does not establish a photonic bandgap or a cavity Q.**

Deliverable: A thickness-width map showing the relevant guided-mode families and their polarization content.

2.2 Simulation B: 3D unit-cell band structure

Purpose: Find the guided stopband used for confinement and see how it changes with fabrication.

Build one truly periodic unit cell with finite beam thickness and realistic surroundings in the transverse directions. Apply Bloch periodicity along the beam:

$$\mathbf{E}(x + a, y, z) = e^{ik_x a}\mathbf{E}(x, y, z).$$

Apply Floquet pairing to the **complete corresponding longitudinal electromagnetic faces, including air**, with compatible source/destination meshes [35]. Use an open transverse treatment; any transverse PML and its mesh must be compatible with the longitudinal periodic mapping. Inspect face selections after each geometry regeneration.

Pick one normalization and use it throughout:

$$q = \frac{k_x a}{\pi}, \quad q \in [0, 1] \qquad \text{or} \qquad \tilde{k} = \frac{k_x a}{2\pi}, \quad \tilde{k} \in [0, 0.5].$$

Start with about 25-40 k-points, refining near band edges and avoided crossings.

For every branch, store the frequency, field profile, how much of the field sits in diamond, and the polarization character. Track modes by field overlap between neighbouring parameter points. Do not just call the third eigenvalue "band 3".

A suspended nanobeam has a **guided-mode stopband**, not a complete 3D photonic bandgap. Radiation channels remain, and symmetry breaking can couple the target mode into other channels [4].

> **Important distinction:** A real tapered cavity is not globally periodic. Its "local band structure" is a diagnostic: you repeat a representative local cell periodically. It is not the exact band structure of the whole fabricated device.

Deliverable: Band diagrams for the mirror cell, the central cell and representative etched versions of both.

2.3 Simulation C: complex eigenfrequency of the finite cavity

Purpose: Get the actual cavity resonance, vector field and total electromagnetic decay rate.

Build the full cavity, without roughness at first. Include the taper, mirrors, output section and any nearby substrate or support that can affect leakage.

With the convention $e^{-i\omega t}$:

$$\tilde{\omega} = \omega_c - i\frac{\kappa}{2}, \qquad Q = \frac{\omega_c}{\kappa}.$$

Use absorbing boundaries/PMLs outside the physical domain. A complex-eigenfrequency analysis gives resonant modes and their losses directly. Numerical modes and PML modes must be filtered out carefully [3].

> **Solver convention note.** The project convention is $e^{-i\omega t}$. **COMSOL uses** $e^{+i\omega t}$, so a decaying positive-frequency mode has the opposite sign of $\operatorname{Im}\tilde{\omega}$ there. Record each solver's phasor convention, and convert all exported complex frequencies, fields and loss parameters to the project convention before comparing solvers.
>
> For the passive, positive-frequency branch:
>
> $$Q = \frac{\operatorname{Re}\tilde{\omega}}{2|\operatorname{Im}\tilde{\omega}|}, \qquad \kappa = 2|\operatorname{Im}\tilde{\omega}|.$$
>
> Keep and check the original sign. Taking the absolute value must not hide an unphysical growing solution or an error in the source/material convention.

Frequency-unit contract. Keep the solver's raw complex frequency and units. For an export $\tilde{f}$ in Hz, use $\tilde{\omega} = 2\pi\tilde{f}$, so $\kappa = 4\pi|\operatorname{Im}\tilde{f}|$. Explicitly distinguish g from $g/(2\pi)$, and energy decay from amplitude decay. Convert complex fields and source/material conventions consistently when changing phasor convention.

For dispersive materials, apply the nonlinear or self-consistent procedure in Section 1.1 to both finite-cavity and band eigenproblems. Recheck the energy/coupling normalization if dispersion is no longer weak.

For each geometry, save:

Quantity	Why it matters
Resonance wavelength	Spectral targeting and tuning requirements
Total Q	Overall photon lifetime
Full complex E_x, E_y, E_z	Dipole coupling and polarization
Mode localization	Detecting mode switching or delocalization
Stored-energy distribution	Identifying sensitive boundaries
Nearby resonances	Detecting multimode behavior
Geometry and mesh metadata	Reproducibility

Do not apply symmetry boundaries after introducing a distortion that breaks that symmetry.

2.4 Simulation D: driven cavity and output-mode decomposition

Purpose: Separate useful output from unwanted losses.

A large Q does not tell you whether photons leave through the desired waveguide.

Run a driven frequency-domain calculation or a time-domain excitation. Decompose the outgoing fields into the actual waveguide modes. Integrating the total flux through an end plane is not enough when higher-order modes or radiation are present [5].

Split the total loss rate into channels:

$$\kappa = \kappa_{\text{target}} + \kappa_{\text{other guided}} + \kappa_{\text{radiation}} + \kappa_{\text{absorption}}, \qquad \eta_{\text{out}} = \frac{\kappa_{\text{target}}}{\kappa}.$$

Default extraction: fit the complex resonant scattering response. In the frequency-domain template, fit the complex multiport amplitudes over a resolved resonance window with a passive, consistently normalized single-mode coupled-mode model that includes a smooth direct background [34]:

$$s_{\text{out}} = s_{\text{background}} + c_{\text{out}} a.$$

Power contains interference between the direct and resonant terms. An on-resonance transmission or raw driven-power ratio is therefore not automatically $\kappa_{\text{target}}/\kappa$. Use enough independent excitations/ports to identify the partial rates; record fit residuals, parameter uncertainty and reference planes. If partial rates are not identifiable, use the energy/flux or ringdown method rather than reporting an arbitrary fit decomposition.

Independent check: source-off ringdown. Excite the selected resonance, switch the source off, let prompt transients leave and fit a time interval dominated by that mode:

$$U(t) \propto e^{-\kappa t}, \qquad P_i(t) = \kappa_i U(t).$$

Use the same physical stored-energy region (excluding PML), modal channel definitions and reference planes. Account for propagation delay/loss between the cavity and flux planes. Check stability against the fitted time window; beating or nonexponential decay requires a multimode treatment. A justified isolated-resonance energy/flux calculation is also acceptable, but raw driven stored energy may include a background.

Require nonnegative channel rates and agreement of $\sum_i \kappa_i$ with the complex-eigenfrequency decay rate within the Section 2.6 tolerance. Separately determine absorption where present; unresolved nonguided loss cannot be labeled radiation without that separation. Record the extraction method for every loss budget. The

full driven-dipole calculation in Section 2.5 keeps direct emitter radiation and interference; do not replace it with the cavity-only leakage fraction.

If the output taper is simulated separately, define a clear reference plane and multiply by its transfer efficiency only once.

Deliverable: A loss budget split by channel for every finalist, not just a total Q.

2.5 Simulation E: emitter coupling and emission

Use two levels of modeling.

Screening level: overlap with the cavity mode

For an isolated, high-Q, weakly dispersive mode, define an effective mode volume specific to the emitter:

$$V_j(\mathbf{r}) = \frac{\int \epsilon_r(\mathbf{r}')|\mathbf{E}(\mathbf{r}')|^2 \, d^3r'}{n_d^2 \left|\hat{\mathbf{d}}_j^\dagger \mathbf{E}(\mathbf{r})\right|^2}.$$

This combines field intensity, emitter position and dipole alignment. It is more useful than quoting only the mode volume at the global field maximum.

Use Q/V_j, or a consistently normalized g_j^2/κ, to screen candidates. For open resonances, this stored-energy normalization is a high-Q approximation, so check the final coupling with driven calculations.

Normalization and integration domain. The implementation must declare:

- **The physical integration region**, with a convergence test as that region grows. Do not let the stored-energy integral run over PML cells, and do not integrate only over diamond while silently leaving out the physical field in air.
- **What kind of field is saved:** unnormalized eigenvector, energy-normalized field, or field from a specified source power. These are not interchangeable, and the saved metadata must say which one it is.

For finalists, keep the driven-emission validation below rather than relying only on the eigenmode normalization.

Validation level: dipole emission into the full structure

Place an electric dipole at the chosen emitter position. Compute the emitted power and how it splits between target and unwanted channels. Normalize against the **same dipole in homogeneous diamond**, not vacuum. Classical dipole/LDOS calculations give the electromagnetic enhancement factors needed to model spontaneous emission [6].

For an emitter that starts in the excited state, a useful rate-model endpoint is:

$$P_{\text{use}} = \eta_{\text{transfer}} \frac{\Gamma_{C,\text{target}}}{\Gamma_{C,\text{total}} + \Gamma_{D,\text{total}} + \Gamma_{\text{PSB}} + \Gamma_{\text{nr}}}.$$

The target contribution is already part of $\Gamma_{C,\text{total}}$, so do not count it twice. Evaluate each optical transition at its own frequency.

Tracking each transition separately matters for SnV, because C and D have different dipole orientations and branching contributions [7].

Rate bookkeeping. Declare bulk lifetime, quantum efficiency, Debye-Waller factor, C/D branching weights and their evidence/uncertainty in `emitter_spec.json`. Also record operating temperature, excited-state branch populations, homogeneous linewidth (or population decay and pure dephasing separately), slow spectral-diffusion distribution/correlation time, and tuning measurement/update times. The C/D-only model requires the relevant lower excited-state branch to be populated; include other transitions when that approximation fails [7]. The C, D, phonon-sideband and nonradiative bulk rates must sum to the stated bulk population-decay rate. Use

$$\Gamma_{C,\mathrm{total}} = \Gamma_{C,\mathrm{bulk}} F_{C,\mathrm{total}}, \qquad \Gamma_{C,\mathrm{target}} = \Gamma_{C,\mathrm{bulk}} F_{C,\mathrm{target}}.$$

Each F is a power ratio relative to the same source in homogeneous diamond. The total factor already includes background emission; do not add a second bare C rate. Model D at its own frequency. If the sideband rate remains at its bulk value, label and test that approximation where broadband LDOS changes matter. Near interfaces, check source-size/mesh convergence and point-source self-field treatment.

Operating-regime and linewidth gate

The branching-ratio endpoint assumes effectively Markovian, competing channels with constant rates. A monochromatic dipole solve measures the electromagnetic response at one frequency; it does not include the emitter's finite optical linewidth automatically.

For an isolated transition coupled to one mode, define g, detuning Δ, and all rates in angular-frequency units. Let Γ_{bg} be population decay excluding this explicit cavity channel, and γ_ϕ the pure coherence-decay rate. In an incoherent exchange approximation [33],

$$K = \kappa + \Gamma_{\mathrm{bg}} + 2\gamma_\phi, \qquad R(\Delta) = \frac{4g^2}{K} \frac{1}{1 + (2\Delta/K)^2}.$$

Thus $R(0) = 4g^2/K$ includes homogeneous broadening. It is an exchange rate; treating it as an extra one-way decay channel requires an appropriate bad-cavity limit. For negligible cavity back-transfer, require $R/\kappa \ll 1$ and validate elimination of the cavity dynamics. The source convention in [33] uses a dephasing parameter equal to $2\gamma_\phi$ here.

At 620 nm, the cavity intensity FWHM is $\kappa/(2\pi) = c/(\lambda Q)$:

Cavity Q	$\kappa/(2\pi)$
10^5	4.84 GHz
10^6	484 MHz
3×10^6	161 MHz

Required check: record g/K, R/κ, $\Gamma_{\mathrm{bg}}/\kappa$ and the emitter-to-cavity linewidth ratio for the operating range. Compare the rate prediction with a single-excitation master equation for representative nominal, broadened and detuned cases; require agreement within the declared probability tolerance (initially 3% relative, tighter than any claimed gain). If coherent exchange, back-transfer, branch dynamics or non-Markovian fluctuations invalidate the reduction, use the fuller dynamics for affected candidates.

Avoid double counting: choose either full electromagnetic LDOS factors or an explicit cavity-plus-background model. Do not add R to a rate that already includes the same cavity resonance. Establish the noncavity background and account for shared-channel interference when using an explicit mode.

For spectral diffusion slow compared with one emission event, average the resulting P_{use} over **residual detuning after the declared tuning procedure**. Do not average a rate first and insert it into the nonlinear branching ratio. Fluctuations faster than tuning updates cannot be perfectly retuned for each shot; fluctuations during emission need a suitable stochastic or dephasing model. Section 3, Step 8 specifies the tuning order.

When the rate model fails this gate, use a cavity-emitter master equation for the finalists. The target photon probability is then:

$$P_{\text{target}} = \kappa_{\text{target}} \int_0^\infty \langle a^\dagger a \rangle(t)\, dt.$$

This counts the cavity contribution to the target channel for an initially excited emitter without continued pumping. If direct emitter radiation into the same output mode is appreciable, use the full channel response, including interference where relevant. This alone does not establish photon indistinguishability or spin-photon entanglement fidelity.

2.6 Validate the numerical framework before optimizing

- Use at least three mesh resolutions, and vary the physical-domain size and PML configuration independently.
- Proposed acceptance targets for finalists: under ~5% change in Q, a few percent in coupling, and a few percentage points in output efficiency. Tighten them whenever the claimed design improvement is of similar size.
- Judge wavelength accuracy against the **tuning margin and the comparison being made**. A solver that looks stable to 0.1 nm cannot establish picometer-level absolute spectral targeting.
- Check selected geometries with a second solver or a time-domain ringdown. Check power conservation in driven calculations.

> **Milestone:** Reproduce a published baseline's optical behavior and explain any remaining discrepancy before modifying the design.

Numerical acceptance record

These are proposed initial tolerances, not universal accuracy guarantees. Refine mesh, domain and PML independently; compare the same tracked mode against the finest solution.

Check	Initial acceptance target
Mesh, domain and PML	Three mesh levels; independent domain/PML changes; less than 5% change in Q
Emitter coupling	Less than 3% change in the selected normalized coupling metric
Output efficiency	Less than 0.02 absolute change in η_{out}
Channel-rate extraction	Nonnegative rates; $\sum_i \kappa_i$ agrees with eigenfrequency κ within 5%; stable fit/window
Emitter reduction	Rate and master-equation photon probabilities agree within 3% relative over declared operating cases
Resonance	Predeclare a wavelength tolerance tied to tuning resolution and the detuning comparison
Driven power balance	Less than 1% residual between source/input power and all outgoing plus absorbed power, normalized to source/input power
Mode identity	Consistent localization, polarization and overlap; inspect nearly degenerate subspaces

Tighten tolerances when a claimed gain is comparable to numerical uncertainty. Save `baseline_validation.csv`, field slices and accepted solver settings. Unresolved mode switches or power-balance failures prevent promotion of the affected model. Explain differences in geometry, materials, ports and normalization in the published comparison; do not fit arbitrary absorption to force agreement with measured Q.

3. Workflow for Idea 1: process-aware, undercut-tolerant design

Step 1: Parameterize the mask and release layout

Start with about 8-12 optical design variables rather than optimizing every hole independently. A suitable set: mirror period, central period, beam width, mirror-hole dimensions, two or three taper coefficients and output-mirror strength. Keep mirror and taper cell counts fixed at first, and explore those discrete choices in a separate outer loop.

Add only a few release-layout variables, such as side-trench distance, release-access width, or a smooth variation of access spacing along the cavity.

Class	Examples
Design controls	Periods, hole contours, beam width, release openings
Process controls	Protected depth d_1, extra depth d_2, intended undercut progression
Uncontrolled variation	Mask bias, etch-rate variation, asymmetry, roughness

> **Critical constraint:** In a bulk-undercut model, final thickness and bottom shape are generally *outputs* of the process. You cannot set both freely without regard to how far the etch has progressed.

Step 2: Build a reduced-order fabrication model

Use the process sequence itself as the order in which geometry is generated:

$$\text{transfer to } d_1 \rightarrow \text{sidewall protection} \\ \rightarrow \text{breakthrough} \rightarrow \text{extra etch } d_2 \rightarrow \text{release.}$$

Start with a parameterized model, not a full plasma simulation.

Construct the pre-undercut solid and protection map explicitly. Define d_1 as the diamond etch depth when sidewall protection is deposited, and d_2 as the additional directional diamond etch depth after horizontal-protection breakthrough. Deposit protection after d_1, open the horizontal protection, then extend the exposed diamond etch by d_2. Upper sidewalls remain protected while the newly created lower sidewalls are exposed. Save the solid and surface labels at every step; use $d_2 = 0$ when the chosen recipe omits the extra etch.

Mouradian et al. [31] report an additional directional oxygen etch of about 1 μm after alumina breakthrough. The Sawfish fabrication process [32] also creates exposed lower sidewalls, using its own protection material and sequence. These examples define possible initial conditions, not universal depth values. Record the actual protection material, thickness, coverage and breakthrough behavior. Protecting every sidewall down to the final trench floor would describe a different release geometry.

Model A: cross sections that vary along the beam

Describe the underside with a few parameters: residual thickness, facet/curvature amplitude, lateral asymmetry and variation along the beam. Treat sidewall angle and pattern-transfer bias separately.

Parameters in neighbouring cells must be correlated. One etch realization should produce one coherent device, not a set of unrelated cross sections.

Model B: geometric surface evolution

As an independent check, evolve the exposed diamond surface with a normal etch velocity that depends on orientation. Protected surfaces stay protected until the model explicitly removes or exposes them.

Experiments show that diamond dry etching depends on crystallographic orientation [8], so a single isotropic erosion rule will not fit every recipe.

Validate the process model itself:

- no diamond is created
- no etching through permanently protected surfaces
- no artificial access to sealed regions
- results converge as the surface discretization is refined

Initial process-model contract: exposed-surface evolution

Use Model A for sensitivity studies. Release-layout optimization requires an explicit Model B. The following is a **proposed scenario model**, not a measured diamond etch law. Reference [8] motivates orientation dependence, but does not supply the access law or coefficients below.

Inputs: mask and release openings, d_1, d_2, protection material/coverage and breakthrough rules, sidewall angle/bias, crystal-to-device rotation, shared etch progression and a correlated process realization. Generate the protection labels by the sequence above; labels follow the protected interface, not a fixed grid location.

Start after breakthrough and the declared d_2 etch, including the bulk floor, exposed trench bottoms and any exposed lower sidewalls. Let $D(s)$ denote remaining diamond and s nominal etch distance in nm. Move accessible, unprotected surface points inward by

$$\Delta \ell = \Delta s \, A(\hat{\mathbf{n}}_{\mathrm{crys}}) \, T(\mathbf{r}, s), \qquad \Delta s > 0.$$

Transform the surface normal into crystal coordinates. An initial smooth cubic-symmetry scenario is

$$A(\hat{\mathbf{n}}) = \exp\left[\alpha \left(n_1^4 + n_2^4 + n_3^4 - \frac{3}{5}\right)\right].$$

The dimensionless α controls anisotropy; $\alpha = 0$ is isotropic. This positive law respects cubic symmetry but imposes a restrictive rate ordering:

$$A_{100} = e^{2\alpha/5}, \qquad A_{110} = e^{-\alpha/10}, \qquad A_{111} = e^{-4\alpha/15},$$

$$\ln \frac{A_{100}}{A_{110}} = 3 \ln \frac{A_{110}}{A_{111}}.$$

The $\langle 110 \rangle$ rate is always intermediate unless all three are equal. Fitting α cannot represent arbitrary relative speeds. Use this family to test the pipeline and limited scenarios; [8] establishes recipe-dependent faceting, not these coefficients.

Orientation/layout recommendation gate: verify normal-speed ordering against the selected recipe and repeat the comparison with a more flexible angular law, such as additional independent cubic invariants or a symmetry-consistent tabulated/interpolated speed function. If facets drive the optical result, the alternative law

must reproduce the relevant facets. Agreement within this one-parameter family alone cannot support a general crystallographic optimum.

For an optional access-limited scenario, use

$$T(\mathbf{r}, s) = \frac{1}{1 + \beta L_{\mathrm{access}}(\mathbf{r}, s)/L_{\mathrm{ref}}}, \qquad \beta \geq 0.$$

L_{access} is the shortest path through connected void from a declared external etchant-entry boundary to the exposed surface. Set $T = 0$ on protected or inaccessible surfaces. Permanent protection is a barrier in this accessibility calculation. Recompute connectivity and paths as the front moves. Keep the reference length L_{ref} fixed across designs.

Release spacing and opening width affect the starting solid, exposed boundaries and access paths. This simple path law does not resolve diffusion, depletion, plasma chemistry or full aperture-dependent transport. Include $\beta = 0$ as a transport-free control. Claim no width-transport benefit if changing width changes none of the inputs controlling the front.

Implementation sequence:

1. Apply mask bias and transfer to d_1; deposit protection, open horizontal protection, and extend the directional etch by d_2. Export the resulting protected/exposed surface map.
2. Advance reachable, unprotected interfaces with a level-set/front method or equivalent geometric erosion; update access each step.
3. Save solids at shared s values. Derive thickness, underside shape, substrate gap and release status from the solid rather than optimizing them independently.
4. Export solids and cross-section summaries. Reject disconnected beams or minimum-feature violations. Incomplete release may be simulated diagnostically but fails any criterion requiring release.

Converge step size and surface resolution, including the release transition. Require $D(s + \Delta s) \subseteq D(s)$ within the recorded discretization tolerance. Check a planar front, a protected surface, a sealed void and symmetric two-sided release. Any symmetry breaking must come from an input or quantified numerical error.

First comparison matrix: isotropic/unrestricted access; orientation-dependent/unrestricted access; orientation-dependent/access-limited scenarios. Hold mask and process samples fixed. Introduce randomness through declared inputs; any extra positive spatial rate multiplier requires its own correlation model. Do not refit coefficients separately for each candidate design.

Set progression points using a common reference recipe. Stopping each device at its best thickness assumes per-device endpoint control and must be evaluated separately. Without calibration, s is equivalent distance, not laboratory etch time.

Model A may later approximate Model B within a validated domain. Their agreement is numerical consistency, not independent physical validation. Use independent data or a separately justified model family for the final robustness check.

Calibration data (needed for a calibrated claim, not to start)

The simulation study can begin without experiments. A **calibrated** process-aware claim needs a small geometry dataset rather than a large optical experiment:

- top-view SEM, plus cross-sectional or suitably tilted SEM
- thickness or topography measurements where accessible
- a verified crystal-coordinate reference
- controlled variation of beam width, release spacing, orientation and etch progression

A top-view image alone does not uniquely determine the underside profile.

For a simulation-only paper: Without calibration, call the variable "etch progression" rather than assigning unverified minutes. Report robustness across plausible families of shapes, not a predicted laboratory yield.

Deliverable: For one mask, a family of physically allowed geometries at early, nominal and late release.

Step 3: Build the uncertainty model

Define global and local variation separately:

$$\mathbf{p}_{\mathrm{fab}} = \mathcal{G}(\mathbf{m}, \mathbf{l}, \mathbf{u}, \boldsymbol{\xi}),$$

where $\mathbf{u}$ holds the shared process variables and $\boldsymbol{\xi}$ holds the residual spatial disorder.

Variation	Correlation structure
Global mask scale or etch bias	Shared across the whole device
Depths d_1 and d_2	Declared joint distribution; global and slow spatial components
Undercut progression	Shared variable affecting many cross sections
Left/right release imbalance	Correlated asymmetry
Bottom-shape variation	Correlated between neighbouring cells
Surface roughness	Defined RMS amplitude and correlation function
Material index uncertainty	Global optical shift

Do not treat a signed boundary displacement as separate, independent width and hole-radius errors. For example, a shared diamond-erosion bias narrows the beam *and* enlarges the holes.

Where there is no data, use explicitly labelled uncertainty scenarios (for example conservative and aggressive) and test several distributions. A Gaussian prior is an assumption, not a measured fact.

Deliverable: A versioned uncertainty specification, used identically for every design comparison.

Store distributions, bounds, correlations, coordinate systems and seeds in `process_spec.json`. Mark inputs as measured, fitted or assumed. Sample latent process variables and then generate the whole device. Do not add independent thickness noise after the etch model has already generated thickness. Reserve separate seeds for training, model selection and final testing.

Step 4: Find out why the baseline fails

Before optimizing, find out what actually limits the baseline.

Run controlled sweeps of thickness, underside shape, sidewall angle, gradients along the beam and asymmetry. These one-variable sweeps are diagnostic. The final statistical assessment uses the correlated process model.

For each sweep, compute the resonance wavelength, Q, coupling to the target mode, and how the local band edges move.

Failure mechanism	Diagnostic signature
Common spectral drift	Resonance moves, but confinement and field shape stay similar
Differential band-edge drift	Mirror and defect cells shift by different amounts
Radiation increase	Transverse loss rises with no matching gain in useful output
Polarization mixing	New field components, or coupling to another mode family
Output mismatch	Cavity stays confined but transfer into the target mode gets worse
Incomplete release	Leftover connections or nearby substrate affect the mode

The finite undercut distance matters. Compare an ideal free-standing cavity with a geometry that includes the real nearby bulk-diamond floor.

Chen et al. discuss underside nonuniformity caused by the quasi-isotropic etch [9], which gives direct experimental motivation for this analysis.

> **Go/no-go criterion:** Identify at least one substantial failure mechanism that correcting the mean dimensions does not remove.

Step 5: Track confinement along the etch trajectory

This should be the main physics component.

For representative cells j, compute how their guided band edges move with etch progression s, and track the cavity resonance relative to those edges.

Separate common spectral drift from differential motion of the cavity and the relevant confining mirror edge. One candidate diagnostic is

$$S_{\mathrm{diff}}(s) = \frac{\partial}{\partial s}[\omega_c(s) - \omega_{\mathrm{edge,mirror}}(s)].$$

Track the same modes/edge and use converged finite differences at shared progression values. Test whether a realizable mask or release-layout change reduces this differential sensitivity over a useful interval while preserving output coupling. This is a mechanism to investigate, not a guarantee of high Q: radiation, finite-mirror attenuation and output mismatch still require full-device checks.

Where useful, estimate the local evanescent attenuation:

$$\chi_j(s) = a_j \,\mathrm{Im}\, k_{B,j}[\omega_c(s), s].$$

A real-k band calculation does not give this imaginary wavevector directly. Get it from a complex-k formulation or a driven calculation with a finite mirror.

For the driven approach, simulate several mirror lengths and extract the attenuation from the internal field decay or from how transmission changes with length. Leave out regions dominated by interfaces and the numerical transmission floor.

Deterministic cavity-design theory relates mirror strength to nanobeam confinement [10], and Sawfish already discusses fabrication-induced band motion and loss of the cavity condition [11]. The proposed contribution is narrower: **test whether a realizable mask or release-layout modification suppresses a particular optical failure caused by correlated undercut geometry evolution**, beyond mean correction and a strong geometric-error optimizer.

Deliverable: A map of band-edge alignment or attenuation against cell position and etch progression, together with full-cavity Q and field profiles.

Step 6: Build the four necessary design controls

Optimize all four within the same architecture and constraints:

Strategy	Optimization assumption
Nominal design	Perfect intended geometry
Mean-corrected design	One deterministic mean fabricated geometry
Independent-error robust design	Independent geometric perturbations
Process-aware robust design	Correlated variations generated by the fabrication model

Evaluate all four on the **same held-out process distribution**. Then repeat the strongest comparison with a second architecture, such as an open-corrugation design.

Match the comparison: use the same available mask variables, geometric constraints, objective, emitter/tuning assumptions and computational search budget. Match the independent-error amplitudes and marginal distributions to the process-generated geometry as far as possible; document residual mismatches. Changing both correlations and error severity would confound the comparison.

Separate two benefits with an ablation: first hold the release layout fixed and compare uncertainty models using identical optical design variables. Then allow the same declared release-layout variables and budget in the applicable strategy comparisons. Report the gain from process information separately from the gain from adding layout freedom.

This is necessary because Sawfish [11] and the 2025 deep-learning-optimized, fabrication-error-tolerant nanobeam cavity of van Haagen et al. [14] already include substantial optimization and uncertainty analysis. Treat [14] as an important baseline, not only a citation.

> **The central comparison is not "optimized versus unoptimized". It is "process-aware optimization versus the best simpler alternative".**

Step 7: Define the robust objective

Choose and freeze the optimization objective after the first mechanism and orientation screens. A lower-tail coupling metric can be used provisionally; once the linewidth and output models are validated, use P_{use}. Predeclare the objective, thresholds and tie policy before the optimization campaign.

For a random performance P, one option is the mean of the worst 10% of realizations:

$$J_{0.1} = \frac{1}{0.1}\int_0^{0.1} F_P^{-1}(u)\,du.$$

This penalizes a design that has an excellent median but a large failure tail.

Also report a yield curve as a function of threshold:

$$Y(P_{\min}) = \Pr(P_{\text{use}} \geq P_{\min}).$$

Reporting the whole curve is better than picking one threshold that happens to favour the proposed design.

Fix the yield denominator. Excluding realizations where the emitter is etched away can make an aggressive geometry look artificially robust. Define and report both:

$$Y_{\text{total}}(P_{\min}) = \Pr[\text{valid geometry, surviving emitter, } P_{\text{use}} \geq P_{\min}],$$

$$Y_{\text{conditional}}(P_{\min}) = \Pr[P_{\text{use}} \geq P_{\min} \mid \text{valid geometry and surviving emitter}].$$

Physically invalid geometries and lost emitters are failures in Y_{total}. Assign them $P = 0$ in the unconditional lower-tail objective too. Use positive success thresholds, or explicitly include validity/survival when evaluating a zero threshold.

Zero-tail policy. These choices imply

$$\Pr(P = 0) \geq 0.1 \quad \Longrightarrow \quad J_{0.1} = 0.$$

Thus 12% and 60% physical failure can both give a zero worst-decile score. The default search policy ranks by $J_{0.1}$ first; tied scores are broken by physical survival probability, then $Y_{\text{total}}(P_{\min})$, then unconditional mean performance. Freeze the positive threshold and numerical/statistical tie tolerances in the objective specification. This gives the optimizer a survival signal on a zero-tail plateau without discarding failures or replacing the objective with a conditional tail. Report all components, including the frequency of the zero tail.

Retry or diagnose numerical failures separately. If N draws contain S confirmed successes and M unresolved numerical cases, the empirical total-yield bounds are

$$\frac{S}{N} \leq Y_{\text{empirical}} \leq \frac{S+M}{N}.$$

These are numerical-status bounds, not statistical confidence intervals. Resolve missing cases or demonstrate that the comparison survives these bounds; never silently drop them.

Step 8: Add finite tuning

For each fabricated realization, optimize only over the tuning settings that are actually available:

$$P^*_{\text{use}} = \max_{v \in \mathcal{T}} P_{\text{use}}(v),$$

where $\mathcal{T}$ describes the physical tuning operation.

With slow residual spectral diffusion δ that cannot be tracked shot by shot, optimize the expected photon probability at an allowed setting:

$$P^*_{\text{use}} = \max_{v \in \mathcal{T}} \mathbb{E}_{\delta|\text{available tuning data}}[P_{\text{use}}(v, \delta)].$$

Specify measurement noise, drift timescale and update cadence. Averaging an independently optimized probability for every unseen fluctuation assumes tracking that may not exist. If feedback can resolve those fluctuations, model the actual measurement-and-control policy instead.

Start by comparing no tuning, a finite tuning budget, and perfect alignment as an upper bound. If tuning can only redshift, use a one-sided window.

Relevant SnV cavity experiments tune by gas deposition/desorption [12]. Keep its achievable range and optical penalties as explicit model inputs rather than assuming universal values.

If coating-based tuning is part of the main claim, model the coating on the real geometry for the finalists and recompute both the resonance and the losses.

Deliverable: Yield versus tuning budget, including the best as-fabricated wavelength offset.

Step 9: Optimize in stages

1. **First:** optimize smooth mean geometries.
2. **Second:** add global process variation.
3. **Third:** add correlated spatial variation.
4. **Fourth:** add surface roughness and uncertainty in emitter position.

Start with a modest parameterization and a derivative-free or surrogate-assisted optimizer. Do not add hundreds of free contour variables until the simpler model shows a real advantage.

When comparing nearby candidates, use the same process samples for each, and refresh the training samples periodically. Keep the final test ensemble untouched.

A surrogate can guide the search, but the best candidates and their failure tails must be checked with full electromagnetic calculations. More surrogate Monte Carlo samples reduce sampling error, not surrogate bias [11].

Step 10: Establish the mechanism and the paper's result

Run diagnostic substitutions: replace only the underside, remove only the variation along the beam, or restore only the original output geometry. These artificial geometries are explanatory controls, not proposed fabrication recipes.

The causal chain to establish:

$$\textbf{process variation} \rightarrow \textbf{specific shape change} \rightarrow \textbf{confinement/output change} \rightarrow \textbf{performance distribution.}$$

A convincing paper would contain:

Figure	Question answered
Geometry/process families	What shapes can the process produce?
Baseline sensitivity maps	Which variations matter?
Local bands and finite-cavity fields	Why does performance degrade?
Four-strategy comparison	Does process awareness beat simpler correction?
Tuning and emitter statistics	Does the benefit survive at the interface level?
Cross-model validation	Is the improvement tied to one error generator?

> **Success criterion:** A meaningful improvement in the lower tail or in yield that holds up under independent process-model assumptions. A smaller percentage drop from ideal Q is not enough.

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4. Workflow for Idea 2: crystallographic and vector-field co-design

Step 1: Define the emitter problem precisely

Case	Appropriate objective
A selected emitter with known orientation	Maximize useful coupling for that transition
Unknown orientation drawn from a population	Improve expected performance, lower-tail performance, or threshold yield

Start with the **known-orientation case**, which is easier to interpret.

Start with a low-strain SnV dipole model, then test angular uncertainty or a fuller transition model where justified. The published C/D analysis puts the C dipole along a $\langle 111 \rangle$ direction and distinguishes it from the polarization of the D transition [7].

Do not assume the optical electric field points along the nanobeam axis. For a conventional TE-like mode, the dominant field is usually transverse, but the calculations must use the actual vector field.

Step 2: Implement the crystal-to-device coordinate transformation

For a beam at in-plane angle ϕ, define its axis and transverse direction:

$$\hat{\mathbf{x}}_b = (\cos\phi, \sin\phi, 0), \qquad \hat{\mathbf{y}}_b = (-\sin\phi, \cos\phi, 0).$$

Transform the crystallographic dipole into beam coordinates, including wafer miscut where relevant. Then compute:

$$\left| \hat{\mathbf{d}}_{j,b}^{\dagger} \mathbf{E}_b(\mathbf{r}) \right|^2.$$

Use the complex field components. The fractions $|E_x|^2$, $|E_y|^2$ and $|E_z|^2$ alone do not determine coupling to an arbitrary dipole, because the relative phases can matter.

Deliverable: A tested coordinate module that returns the dipoles and coupling maps for any beam angle.

Step 3: Run two essential analytical/numerical checks

Check A: invariance under rigid rotation

With scalar permittivity and homogeneous surroundings, rigidly rotating the whole optical structure does not change its resonance or Q.

Use this as a numerical control. Crystal orientation enters through the dipole directions and the shape produced by fabrication, not through an invented anisotropic index.

If an ideally rotated but otherwise identical cavity shows a substantial change in Q, look for problems with mesh orientation, boundaries, ports or inconsistent geometry.

Check B: polarization bounds

For a C dipole along $[111]$, $\hat{\mathbf{d}} = \frac{1}{\sqrt{3}}(1,1,1)$, a purely in-plane unit field satisfies:

$$\max |\hat{\mathbf{d}} \cdot \hat{\mathbf{e}}|^2 = \frac{2}{3}.$$

For four equally weighted tetrahedral axes:

$$\frac{1}{4}\sum_j \hat{\mathbf{d}}_j \hat{\mathbf{d}}_j^T = \frac{1}{3} I \quad \Longrightarrow \quad \frac{1}{4}\sum_j |\hat{\mathbf{d}}_j^\dagger \mathbf{E}|^2 = \frac{|\mathbf{E}|^2}{3}.$$

These make good unit tests and are real scientific constraints.

Selected-emitter headroom. An unrestricted unit polarization reaches overlap 1, while the best in-plane polarization reaches $2/3$. At fixed local field intensity per unit stored energy,

$$\frac{g_{3\mathrm{D}}^2}{g_{\text{best in-plane}}^2} \le \frac{3}{2}, \qquad \frac{g_{3\mathrm{D}}}{g_{\text{best in-plane}}} \le \sqrt{\frac{3}{2}} \approx 1.225.$$

The same $1.5\times$ alignment-only bound applies to Q/V_C at fixed Q. Another in-plane direction provides no extra alignment advantage after the best conventional orientation has been chosen at that point. A redesign can still change normalized intensity, radiation loss, emitter-position tolerance or output coupling.

The photon-probability gain can be smaller: for $\beta = r/(1+r)$ with $r = \Gamma_{\mathrm{cav}}/\Gamma_{\mathrm{bg}}$, increasing r by $1.5\times$ changes β from 0.50 to 0.60 at $r = 1$, or from 0.909 to 0.9375 at $r = 10$ (about 3.1% relative gain), before loss/output penalties.

Equal-population worst case. For four tetrahedral dipoles and a unit complex field, the minimum overlap cannot exceed its mean, $1/3$. Crystal $[100]$ or $[010]$ polarization gives exactly $1/3$ for every orientation, hence

$$\max_{\|\mathbf{e}\|=1} \min_j |\hat{\mathbf{d}}_j^\dagger \mathbf{e}|^2 = \frac{1}{3}.$$

An appropriate conventional in-plane TE polarization already attains this pointwise, orientation-only bound. Improving unknown-orientation performance therefore needs an additional benefit or constraint: a fabrication-favored beam angle, spatial emitter distribution, unequal orientation populations, or changed normalized intensity. This does not make every threshold-yield metric invariant.

> **Rotating the polarization alone cannot raise the orientation-averaged coupling at fixed local field intensity.** It can redistribute coupling, or improve a selected emitter or the threshold yield.

Step 4: Test ordinary beam rotation first

Take the conventional cavity and evaluate several beam orientations, in two passes:

- **Pass A: identical geometry.** Rotate only the dipole coordinates relative to the stored cavity field. With the ideal scalar-material model, no new Maxwell solve is needed for each angle.
- **Pass B: fabrication that depends on orientation.** Regenerate the etched shape for each beam angle, then solve the optical problem again.

Pass A isolates dipole alignment. Pass B shows whether the etch changes which orientation is best.

For each angle, record Q, V_C, useful output coupling and their sensitivity to the process.

> **Go/no-go criterion:** use the $1.5\times$ selected-emitter alignment limit and the $1/3$ equal-population worst-case bound to quantify remaining headroom. Advance local design only with a plausible additional mechanism: better normalized field intensity across emitter positions, alignment under a fabrication-imposed angle constraint, useful E_z with acceptable loss, or improved output/process tolerance. If rotation already achieves the relevant bound and no such mechanism remains, close the feasibility study and retain its result within Idea 1.

Step 5: Map the vector field over a physically allowed emitter region

For the promising orientations, save the full vector field throughout the volume emitters can occupy. Generate maps of

$$|\mathbf{E}|^2, \qquad |\hat{\mathbf{d}}_C^\dagger \mathbf{E}|^2, \qquad |\hat{\mathbf{d}}_D^\dagger \mathbf{E}|^2,$$

and of the useful output probability.

Fix the distribution of emitter positions to a physical fabrication scenario. For implanted emitters, depth is measured from the relevant implantation surface. It is not automatically re-centred in each final beam.

> **Never move the emitter to the new field maximum for each fabrication realization.** That would erase exactly the position sensitivity you want to measure.

Realizations where the emitter's material has been etched away are emitter-survival failures. They count as failures in Y_{total} and are excluded only from $Y_{\text{conditional}}$ (Idea 1, Step 7). Report both.

Deliverable: Coupling maps and statistics over lateral position, depth and dipole orientation.

Step 6: Introduce local geometric controls gradually

Family	Added degrees of freedom	Scientific purpose
Conventional cavity	Global orientation only	Strong baseline
Locally modified cavity	Rotated/reshaped central holes, selected lateral offsets	Reshape the in-plane field direction
Process-aware vector cavity	Local controls plus release-shape controls you can actually set	Exploit the realistic 3D field structure

Start by modifying only a few central cells. Leave the mirrors and output section unchanged until a coupling improvement has been demonstrated.

A key symmetry constraint: changing the hole pattern seen from above does not automatically create an out-of-plane field at a symmetry plane. For a vertically symmetric TE-like mode, parity can force E_z to vanish there. The real released underside may break that symmetry, but its effect must be calculated, not assumed.

Do not optimize bottom facets freely, separately from the process model.

Step 7: Optimize normalized coupling, not polarization percentage

For a selected emitter, the starting objective can be a position-averaged or lower-tail value of

$$\frac{Q}{V_C(\mathbf{r})}.$$

Then replace it with P_{use}, including output coupling and tuning.

This avoids a common false improvement: the field lines up better with the dipole, but its amplitude falls or radiation loss rises enough that useful emission gets worse.

For unknown orientations, compute each orientation separately and combine them according to a stated population. Report whether the improvement comes from stronger local fields, better balance between orientations, or better output coupling. These are different mechanisms and should not be mixed up.

Step 8: Use reciprocity to cut the computational cost

Launch the desired output waveguide mode backward into the structure. Its reciprocal field maps how dipoles at different positions and orientations couple into that output channel [13].

This can replace many separate dipole-to-output calculations, provided the mode-power normalization is handled correctly. The reciprocity formalism applies to emission into a specified propagating mode of an open optical system.

However, it does **not** give the total emission rate.

For representative positions, check the denominator of P_{use} with direct dipole/LDOS calculations.

Three orthogonal dipole simulations at one position can reconstruct the local response for any dipole orientation, **but only if the full complex vector response is kept**. By linearity,

$$\mathbf{E} = \mathsf{G}\,\mathbf{p}.$$

An x-directed source gives one column of G, provided all three field components are recorded. Three independent sources give all three columns.

Saving only the three scalar powers P_x, P_y, P_z generally misses the cross terms. For a dipole along $(\hat{x}+\hat{y})/\sqrt{2}$, a quadratic emission response contains

$$\tfrac{1}{2}(A_{xx} + A_{yy} + 2\,\mathrm{Re}\,A_{xy}),$$

not just $(A_{xx} + A_{yy})/2$.

Requirement: save source-normalized complex vector fields, with a consistent treatment of the point-source self-field. This matters most for asymmetric structures and for Idea 2.

Deliverable: Efficient maps of output coupling across position, cross-checked against explicit dipoles.

Step 9: Reintroduce the full fabrication ensemble

Test the local-field design against:

- Shape variation caused by the undercut
- Sidewall and underside asymmetry
- Mask-angle errors
- Uncertainty in emitter position

- Roughness
- Finite tuning

Small local asymmetries can improve the ideal field but make the design very sensitive to fabrication. The robust comparison must therefore use the **best rotated conventional design**, not an arbitrarily misaligned baseline.

If the modified cavity develops nearby hybridized modes, use the driven electromagnetic response or a multimode model instead of blindly adding isolated Q/V estimates.

Step 10: Decide whether it is a separate paper

Require a useful-emission gain beyond numerical and sampling uncertainty over the best rotated conventional cavity, explained by at least one of these outcomes:

- **Selected-emitter result:** A robust gain in useful coupling over the best conventionally rotated cavity, with a clear explanation of the vector-field mechanism.
- **Orientation-uncertain result:** A meaningful gain arising from a stated constraint or changed field intensity/position tolerance, respecting both the orientation-average and worst-case bounds. Threshold yield can change even when the ideal overlap bound does not.
- **3D result:** A useful out-of-plane field component produced by a released geometry you can fabricate, whose benefits survive position and process uncertainty.

The main figures should show the rotation baseline, vector-field maps, coupling for each dipole, position dependence and the robust comparisons.

If the only result is that a better-aligned field couples more strongly, fold it into Idea 1 rather than forcing a separate publication.

---

5. Managing the simulation workload

5.1 Use a hierarchy of fidelity

Level	Calculation	Role
0	Geometry and connectivity checks	Reject impossible masks cheaply
1	Cross-section modes and local bands	Reject poor regions of optical parameters
2	Smooth full-cavity eigenmodes	Main design exploration
3	Full cavities with process variation	Robust optimization
4	Roughness, output decomposition, dipole validation	Finalist assessment
5	Independent-solver and model checks	Publication-quality validation

Do not run expensive full-device dipole simulations with rough surfaces for every early candidate.

Do not represent roughness as independent displacements of mesh vertices. Specify a physical spatial correlation or power spectrum, then resolve that geometry properly.

5.2 Use staged statistical validation

Use a validation ensemble to select finalists. Keep a separate final test ensemble untouched until designs and decision thresholds are frozen. Repeated selection against a nominally held-out set makes it validation data.

For independent success/failure (Bernoulli) outcomes, the worst-case approximate 95% sampling uncertainty is:

$$\delta Y \approx 1.96\sqrt{\frac{0.25}{N}}.$$

N	Approx. 95% half-width
100	±10 percentage points
400	±5 percentage points

These cover sampling uncertainty only. Use a suitable binomial interval, such as Wilson [30], for actual yield intervals, especially near zero or one. Uncertainty in the fabrication model is separate.

With 100 realizations, the worst-10% objective contains only about 10 tail observations. Report the tail count and an interval for the paired difference. Increase independent final sampling if the interval cannot resolve the claimed improvement. If interim outcomes determine sample size, declare a sequential procedure or reserve a fixed-size confirmation sample.

An illustrative electromagnetic budget: several hundred training geometries, several hundred adaptively selected geometries, and a few hundred held-out realizations per finalist. These are starting budgets and do not guarantee surrogate accuracy.

Count designs, geometry realizations and complete evaluations separately. For a rectangular campaign,

$$N_{\text{geometries}} = N_{\text{designs}}\, N_{\text{process realizations}}.$$

A complete evaluation can include bands/eigenmodes, frequency scans, independent dipole sources, tuning states, mesh/PML refinement and validation checks. Log actual solves and retries as well as unique geometries; multiply by early/nominal/late progression points when those are separate geometries. Benchmark the end-to-end wall time, RAM and output size at each fidelity. Reusing stored fields for compatible positions/orientations saves solves; it does not turn all source-dependent validation into one solve.

Use the same process realizations for competing designs and analyze the paired difference in performance.

5.3 Store enough information to reproduce every result

For each solve, keep:

- design ID and process-realization ID
- units and solver phasor convention
- field normalization type (Section 2.5)
- mask parameters
- process parameters
- roughness seed
- emitter assumptions
- geometry hash
- material model
- solver settings

- mesh statistics
- mode identifier
- extracted metrics
- failure status

Save the final masks and the solids generated by the process model, not only screenshots.

A failed or low-Q realization is part of the dataset. It should not vanish when you make plots.

---

6. Software and implementation plan

Sections 1-5 are a research workflow. This section turns them into an implementation specification: which tool owns each task, how data move between tools, and what custom code each idea needs.

6.1 Tools

Essential starting tools

Task	Tool	Use
Electromagnetic simulation	**COMSOL Multiphysics + Wave Optics Module**	Cross-section modes, Bloch bands, cavity eigenfrequencies, driven fields, dipole excitation, output coupling (mode analysis, eigenfrequency and frequency-domain studies) [15]
COMSOL automation	**MATLAB + LiveLink for MATLAB** (if the license works)	Change parameters, regenerate geometry and mesh, run studies, export results [16]
Sampling and optimization	**Python: NumPy, SciPy; scikit-learn when needed**	Generate fabrication realizations, compute metrics, parameter searches, a first Gaussian-process surrogate [17]
Mask generation and inspection	**gdstk + KLayout**	Generate GDS from design parameters, polygon operations, inspect the mask [18]
Results and reproducibility	**Git, a Python test suite, CSV tables, HDF5 field files**	Track code, test geometry and metrics, store scalar results separately from large field arrays

COMSOL's Optimization Module is not needed. Optimization runs outside COMSOL; COMSOL is the geometry-and-Maxwell evaluation engine, driven through its API [19].

Do not automatically buy both the RF and Wave Optics modules. Both document Floquet-periodic capabilities [20]. If an RF Module seat already exists, first test whether it provides the interfaces and source/port features these models need. Wave Optics is the default for a new optical workflow, not a reason to duplicate an adequate existing license.

Tools to add only when needed

Tool	When it becomes useful	Recommendation
Meep	Independent checks of cavity, transmission, LDOS and output modes	Add after the COMSOL baseline is validated [21]
MPB	Extra periodic band-structure calculations	Optional; COMSOL handles the first Bloch calculations. MPB is a band solver, not a substitute for open-cavity loss calculations [22]
QuTiP	Cavity-emitter dynamics when a rate model is not enough	Only for finalists that need a master equation (Lindblad solver) [23]
ViennaPS	A more developed surface-evolution model for etching	Consider later; it still needs a justified diamond-specific model and calibration [24]

On Windows, Meep's official installation guide points to Linux via WSL rather than a native install [25]. Keep that environment separate from the main COMSOL/MATLAB environment.

Not prerequisites for either paper: a custom FEM solver, a full plasma simulator, a large neural network, or a quantum-dynamics package.

6.2 COMSOL model templates

Turn Simulations A-E into **four reusable COMSOL templates** rather than one enormous model. The mode, periodic, eigenfrequency, port and electric-dipole features they need are documented in the Wave Optics User's Guide [26]. The filenames are proposed project files; none has been built yet.

Template	Simulation	Setup and purpose
`waveguide_cross_section.mph`	A	2D cross section, **Mode Analysis**. Screen width, thickness, sidewall angle and bottom shape.
`unit_cell_bands.mph`	B	3D unit cell, **Floquet periodicity + Eigenfrequency**. Sweep the Bloch wavevector.
`finite_cavity.mph`	C	Full cavity, **Eigenfrequency + open boundaries/PMLs**. Extract resonance, Q and complex vector fields.
`driven_interface.mph`	D, E	Full device, **Frequency Domain** with waveguide-mode or electric-dipole excitation. Extract useful output, other loss channels, emission enhancement.

What stays outside COMSOL (Python or MATLAB)

- **Fabrication ensemble:** choose process parameters and generate correlated geometry parameters.
- **Emitter calculations:** rotate dipoles, interpolate fields at emitter positions, evaluate overlaps, compute rate-model metrics.
- **Robust objective:** combine realizations into quantiles, lower-tail means and yield curves.

This keeps the COMSOL models focused and lets the physics and statistics be tested independently.

Template acceptance includes complete periodic-face pairing with compatible meshes [35], the declared dispersive eigenfrequency procedure [36], and the background-aware scattering fit of Section 2.4. Add a source-off time-domain study only when needed for the independent leakage check; keep its channel definitions consistent with the frequency-domain model.

One geometry specification for simulation and mask

At first, generate parameterized solids directly in COMSOL from a shared parameter specification, and generate the matching GDS with gdstk from **the same specification**.

Never keep one hand-drawn geometry for simulation and a separate one for fabrication. The mask can quietly drift away from the object you optimized.

6.3 Batch-based automation

Avoid a tightly coupled Python ↔ MATLAB ↔ COMSOL call at every optimizer evaluation. Start with batches:

```
Python generates a candidate batch
                ↓
       candidates.csv / JSON
                ↓
MATLAB + LiveLink evaluates the batch in COMSOL
                ↓
       metrics.csv + fields.h5
                ↓
Python checks results and fits/updates the model
                ↓
Python generates the next candidate batch
```

LiveLink supports the parameter, geometry, solver and results operations needed for the evaluation stage [16]. Every record carries the design ID, process-realization ID, units, random seed, solver settings and status (Section 5.3), not just the resulting Q.

Run status is `queued`, `running`, `ok`, `geometry_invalid`, `emitter_lost`, `solver_failed` or `convergence_failed`. Keep geometry, emitter and numerical flags separately even if one status routes the job. Reject duplicate run IDs and concurrent writes to the same output.

Write each run in its own temporary directory. Mark it complete only after output/schema and numerical checks pass. Restart skips validated completed runs with matching hashes. Log each retry and any fidelity change; retries are not independent fabrication realizations.

Fallbacks if LiveLink is troublesome:

- **COMSOL Java API + command-line batch execution.** COMSOL can save models as Java code and run `.mph` or compiled Java model files from the command line [19]. Python manages the input/output files and launches the jobs.
- **MPh**, a community Python interface to COMSOL [27]. Treat it as an optional convenience, not the foundation. It is a wrapper and does not replace COMSOL or its licenses.

Optimization tools, in order

Stage	Tool/method
Initial parameter coverage	SciPy Sobol or Latin-hypercube sampling (`scipy.stats.qmc`)
Diagnostic sensitivity sweeps	Explicit one-parameter and selected two-parameter sweeps
First surrogate	Gaussian-process regression on a modest parameter space
Candidate search	A bounded optimizer applied mainly to the surrogate
Final acceptance	Full-wave evaluation on untouched fabrication realizations

These algorithms already exist [17]; do not reimplement them. Be cautious about running a population optimizer directly against thousands of expensive full-cavity solves. Measure the cost of one validated model first, then set the optimization budget.

6.4 Custom code each idea needs

Idea 1: the process model is the main development task

$$\mathcal{G} : (\text{mask},\ \text{release layout},\ \text{process variables}) \mapsto \text{3D geometry}.$$

Module	Required output
Mask parameterization	Hole contours, periods, beam boundaries, release openings
Process-to-shape map	Thickness, underside profile, sidewall shape and substrate gap along the device
Correlated uncertainty generator	One coherent fabricated geometry per realization
Geometry validator	Connectedness, minimum features, release status, emitter survival
Mode tracker	Consistent identification of the target mode as the geometry changes
Robust-performance evaluator	Lower-tail performance, tuning requirements, Y_{total} and $Y_{\text{conditional}}$

The process-to-shape map must include the release-layout dependence specified in Idea 1, Step 2.

Idea 2: no extra electromagnetic package at first

Idea 2 mainly adds **vector-field postprocessing** to the same COMSOL framework.

Module	Required calculation
Crystal-to-device transformation	Rotate each transition dipole into simulation coordinates
Position sampling	Evaluate the field over a physically defined emitter distribution
Vector overlap	Dipole projection of the complex field, using all complex components
Rotation baseline	Find the best conventionally oriented cavity before local optimization
Local geometry controls	Modify selected central cells within fabrication constraints
Emission validation	Compare overlap predictions with driven-dipole or reciprocal-field calculations

The ideal orientation scan reuses one stored field when geometry is unchanged (Idea 2, Step 4, Pass A), which keeps the first test of Idea 2 cheap. QuTiP is not needed for dipole projections, Q/V_j or simple emission-rate ratios.

6.5 Computing resources: benchmark before buying

- Start on the available workstation. Run a representative finite-cavity model at three mesh resolutions and record **peak RAM, wall-clock time, degrees of freedom and output-file size**.
- A **128 GB RAM workstation** is a reasonable planning target to investigate. It is not a proven requirement until the actual geometry, mesh and solver are benchmarked.
- For large ensembles, compare several independent jobs with moderate core counts against one job using all cores. Choose by measured throughput and memory.
- A **GPU is optional**. COMSOL 6.4 added the cuDSS sparse direct solver, applicable to the linear systems in eigenfrequency analysis [28]. Test its benefit on these models, and keep double precision for the first high-Q validations.
- For cluster runs, check the license first. COMSOL documents Floating Network License requirements for cluster computing [29].

6.6 First implementation milestone

Before installing optional packages or building the full optimizer, one automated batch must pass all of these:

Test	Pass condition
Baseline cavity	Reproducible resonance and Q, with a documented convergence check
Three underside geometries	Geometry regeneration and meshing work without manual repair
Field export	Complex E_x, E_y, E_z, coordinates, units and normalization saved correctly
Orientation postprocessing	Dipole rotation reproduces the analytical bounds (Idea 2, Step 3)
Batch recovery	A deliberately failed case is logged and does not stop the rest of the batch

This implements gates G1-G2 of Section 7 without building the whole campaign prematurely. Start with **COMSOL + MATLAB automation + Python**, then add mask export and independent-solver validation once the baseline is reliable.

6.7 Batch contract and restart behavior

Use one shared schema version across Python, MATLAB and COMSOL. These are proposed project artifacts, not files already delivered by this workflow.

Artifact	Minimum content
`baseline_spec.json`	Frozen geometry, parameter provenance, materials, coordinates and output definition
`process_spec.json`	Model/version, d_1, d_2, staged protection/access maps, angular-law ordering, progression grid, joint distributions and seeds
`emitter_spec.json`	Positions/dipoles, frequencies, bulk rates, temperature/branch populations, linewidth/dephasing, diffusion times/distribution, tuning cadence and validity regime
`candidates.csv`	Batch/run/design/realization IDs, controls, spec hashes, study and fidelity
`metrics.csv`	Every attempted run; units, optical metrics, channel-rate method/uncertainty, linewidth-validity flags and convergence/status
`fields.h5`	Complex fields, coordinates/mesh connectivity, domains, frequency, normalization, source and reference plane
`run_manifest.json`	Code/spec hashes, solver/module versions, mesh/PML settings, resource use, paths and status

Save the objective and zero-tail tie policy with the frozen batch specifications. Keep fit/ringdown traces and residual-detuning assumptions beside the optical metrics.

Use SI units at the solver boundary; convert named nm mask fields explicitly. Save complex data or paired real/imaginary arrays. Interpolate using the mesh/domain labels so an emitter outside diamond is not assigned a coupling.

The batch-status and restart rules are specified in Section 6.3.

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7. What to do first

Work in dependency order. Gates require reviewable evidence, not empty files or a successful solver exit. Assign an owner and measured compute budget when each gate starts. Estimate calendar time after the first benchmark.

Gate	Concrete output	Exit condition
G0: Freeze baseline	Source-traceable cavity; pre-undercut solid and protection map after d_1/breakthrough/d_2; license/API check	No unresolved essential geometry; recipe and protected/exposed surfaces explicit
G1: Validate optics	Modes, mirror/defect bands, cavity resonance/fields; refinement and resource tables	Section 2.6 checks for the studies run; target mode tracked; published comparison explained
G2: Prove batch path	Baseline, three underside geometries and a deliberate failed job	Geometry, meshing, export and restart work; failures stay in the dataset
G3: Diagnose process	Early, nominal and late release at shared progression; angular-law controls; band/resonance/loss diagnostics	Distinguish common drift, differential band motion, radiation and output mismatch; identify a specific failure
G4: Screen orientation	Stored-field rotation scan; orientation-dependent fabricated shapes where needed	Verify $1.5\times$ and $1/3$ bounds; quantify headroom beyond the best rotated conventional cavity
G5: Optimize and confirm	Frozen objective/tie policy; matched four strategies and layout ablation; untouched paired test	Linewidth/leakage gates pass; gain exceeds uncertainty and survives a more flexible, justified process model

First working session: freeze one source-traceable cavity and the actual pre-undercut solid, including d_1, d_2 and protection labels. Recover missing Chen geometry or use the alternative validation route in Section 1.3. Check licenses with an automated toy solve; freeze units and schemas.

First automated batch: one baseline, three declared underside geometries and one intentional failure. Record wall time, peak RAM and status for each. This checks the evaluation path; arbitrary underside shapes do not validate fabrication physics.

First scientific milestone: after validating the optical mode and geometry regeneration, generate early, nominal and late release geometries at shared progression values. Diagnose common wavelength drift, differential mirror-band motion, increased radiation and output mismatch. Reuse the fields for the inexpensive orientation analysis. Then freeze the optimization objective around the limitation that persists after mean correction and realistic tuning.

Idea 1 decision: optimize only if a substantial process-correlated confinement/output failure remains after mean correction. If the change is mainly a common wavelength shift, study mean correction and finite tuning before expanding optimization.

Idea 2 decision: keep a bounded feasibility study until an identifiable mechanism improves useful emission over the best rotated conventional cavity, consistent with the selected-emitter and equal-population bounds. If the benefit disappears under linewidth, position or process uncertainty, incorporate the orientation result into Idea 1.

Final evidence package: frozen specs/masks, solids, convergence and power balance, all attempted runs, coupling/orientation maps, four-strategy comparison, total/conditional yield, uncertainty intervals and model-sensitivity results.

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